

# Lia — Session Summary

The whole build, one line at a time.

Every exchange in the build of Lia, one line for what was asked and a short account of what came of it. Times and tool counts come from Claude Code's session log; the full wording is in *Lia — Session Transcript*.

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|------------------------|--------------------------|---------------------------------|---------------------------|
| <b>50</b><br>exchanges | <b>725</b><br>tool calls | <b>01 Aug, 12:52</b><br>started | <b>~26 hrs</b><br>elapsed |
|------------------------|--------------------------|---------------------------------|---------------------------|

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## Getting her talking

- |                                       |                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b><br>01 Aug, 12:52<br>10 tools | <b>Introduced the project; stuck because the editor's Run button can't take typed input.</b><br>Confirmed the code was fine by running it end to end. The fix was a real terminal. Flagged that DB_PATH was relative, so launching from the wrong folder would silently create a second, empty memory.                         |
| <b>2</b><br>01 Aug, 13:26<br>24 tools | <b>Typed something and nothing came back.</b><br>Not broken — <b>silent</b> . Replies were fetched whole before printing, so ~20s passed with a blank screen. Added token streaming and progress markers. Also found fact extraction was storing nothing, and made DB_PATH absolute.                                           |
| <b>3</b><br>01 Aug, 13:41<br>34 tools | <b>Rename Aurora to LIA; add voice, with room to add my own voices later.</b><br>Built the whole voice layer: Piper for speech, a voices/ folder you drop .onnx files into, sentence-chunked speech so she starts talking before the reply finishes, and Whisper for listening.                                                |
| <b>4</b><br>01 Aug, 13:58<br>10 tools | <b>Fix the two things you flagged.</b><br>Diary was narrating as if it were the human — caused by user:/assistant: labels. Relabelled to <b>Them/Lia</b> . Added a deterministic filter so junk like name: "you" can't overwrite real facts.                                                                                   |
| <b>5</b><br>01 Aug, 14:03<br>5 tools  | <b>Analyse the project.</b><br>Found the biggest quality flaw: <b>half her recall was her own past replies</b> , unlabelled, which she read back as fact. Also: diary written but never read, no sense of time, embeddings stored as JSON (10× bloat). Measured the scan cost and said plainly it wasn't worth optimising yet. |
| <b>6</b><br>01 Aug, 14:06<br>2 tools  | <b>Fix line 8, column 16.</b><br>Type checker couldn't verify sys.stdout.reconfigure. Switched to getattr, which is also more honest about why the guard exists.                                                                                                                                                               |
| <b>7</b><br>01 Aug, 14:10<br>5 tools  | <b>Is everything fine? Explain the directories.</b><br>All modules compiled and ran. Caught her <b>confabulating</b> in that very run — inventing that you'd been nervous. Mapped the folder layout and the strict layering.                                                                                                   |
| <b>8</b><br>01 Aug, 14:13<br>1 tools  | <b>How do I run it?</b><br>The command, what to expect, and why to type <i>bye</i> rather than closing the window.                                                                                                                                                                                                             |

## Giving her a voice

- 9**  
01 Aug, 14:22  
10 tools
- Set it up so I can talk to her by voice.**  
Upgraded to `small.en` after testing showed `base.en` heard “*Ania*” for “*Anaya*”. Biased recognition using names she already knows, which fixed her own name coming out as “*Leah*”.
- 
- 10**  
01 Aug, 14:31  
2 tools
- What if she were an app I just open my laptop to?**  
Measured the real gap: cold start 9.3s vs 2.4s warm. Laid out five things standing in the way and staged them, flagging that **sessions never ending** would mean an always-on Lia never learns.
- 
- 11**  
01 Aug, 14:39  
26 tools
- Build stage 1.**  
Models kept warm, sessions that close themselves after 12 quiet minutes, launcher and autostart script. Found the hidden bottleneck: Ollama's default 32k context was running **63% of her on CPU**. Pinning 8192 cut reply time by a quarter.
- 
- 12**  
01 Aug, 15:54  
31 tools
- Do the retrieval fix, then stage 2.**  
Retrieval limited to your words with timestamps. Added open-mic listening with Silero VAD, keyboard-vs-mic arbitration, and mic gating. Fixed her reading stage directions aloud.
- 

## Turning her into an app

- 13**  
01 Aug, 16:03  
14 tools
- Make her an app I can add to startup.**  
Tray app with no window — only possible because voice detection removed the need for a keyboard. Verified quitting still writes her diary first.
- 
- 14**  
01 Aug, 16:15  
49 tools
- Add a wake word, an exe, and tell me how to start.**  
Wake word matched **on the transcript** rather than a trained model — no dataset needed, and it tolerates Whisper's spellings. Built the exe. Made paths survive being frozen.
- 
- 15**  
01 Aug, 19:57  
1 tools
- How do I get her into startup?**  
The one command, and what it actually does.
- 
- 16**  
01 Aug, 19:58  
1 tools
- If I restart now, will she be there?**  
**No** — I'd built the installer but deliberately not run it. Said so plainly.
- 
- 17**  
01 Aug, 20:00  
5 tools
- Do it.**  
Installed. Hit a real bug of my own: PowerShell variable names are case-insensitive, so my local `$exe` collided with the `-Exe` switch. Verified by launching the shortcut.
- 
- 18**  
01 Aug, 20:03  
1 tools
- Confirm she'll be there after a restart.**  
Yes — both she and Ollama were in the Startup folder.
- 

## Making her yours

|                           |                                                                                                                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>19</b>                 | <b>Greet me, call me Warlock, and can she learn things like Morse code?</b>                                                                                                                                                                     |
| 01 Aug, 20:05<br>17 tools | Greeting generated from her diary, which finally made the diary useful. Tested Morse honestly: <b>she fails even with a perfect reference table in front of her</b> . Not a memory problem — tokenisation. The fix would be code, not teaching. |
| <b>20</b>                 | <b>A folder for PDFs, and make her voice softer.</b>                                                                                                                                                                                            |
| 01 Aug, 20:19<br>36 tools | Built the library with page-level citation. Stripped repeated page headers that were poisoning every search. Played four voice options rather than guessing.                                                                                    |
| <b>21</b>                 | <b>Amy as she was, and make sure she says Warlock.</b>                                                                                                                                                                                          |
| 01 Aug, 20:54<br>10 tools | Reverted the softening. Found the name had <b>reverted on its own</b> — extraction re-derived the old one from stored turns. Added protected facts and purged the stale references.                                                             |
| <b>22</b>                 | <b>Thanks.</b>                                                                                                                                                                                                                                  |
| 01 Aug, 21:01<br>0 tools  | —                                                                                                                                                                                                                                               |
| <b>23</b>                 | <b>Can I say “Lia stop” to stop her listening?</b>                                                                                                                                                                                              |
| 01 Aug, 21:04<br>1 tools  | <b>No.</b> It would just become conversation. And she can't hear you at all while speaking. Said so rather than implying it half-worked.                                                                                                        |
| <b>24</b>                 | <b>What do you mean?</b>                                                                                                                                                                                                                        |
| 01 Aug, 21:08<br>0 tools  | Re-explained without jargon: what barge-in is, why speakers break it, why headphones don't.                                                                                                                                                     |
| <b>25</b>                 | <b>I use both.</b>                                                                                                                                                                                                                              |
| 01 Aug, 21:09<br>18 tools | Built spoken commands, and made interrupting work on speakers too — <b>she compares what she hears against what she's currently saying</b> , so her own voice is recognised and ignored.                                                        |

## Fixing what the real world broke

|                           |                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>26</b>                 | <b>She didn't respond after a restart, and the diary won't open.</b>                                                                                                                                                                                             |
| 01 Aug, 21:34<br>33 tools | Three causes: she spent <b>four minutes re-reading a PDF</b> before speaking (absolute paths made it look new), speech recognition loaded after the greeting, and ignored speech was logged nowhere. The diary was fine — there was simply nothing to write yet. |
| <b>27</b>                 | <b>I didn't want her going through PDFs on her own. And remove Anaya and Kamala.</b>                                                                                                                                                                             |
| 01 Aug, 22:13<br>12 tools | Made the library strictly on-demand. Removed all of it — and said plainly that <b>I had invented Anaya, Kamala, Maya and Biscuit during my own testing</b> . Undid my overreach of having her read your PDF unasked.                                             |
| <b>28</b>                 | <b>The tray boxes were unchecked and she's not responding.</b>                                                                                                                                                                                                   |
| 01 Aug, 22:33<br>18 tools | She'd asked how your day was, then ignored “hello” — my overcorrection from the previous fix. Gave the greeting a 30s reply window. The unchecked boxes were real: the Ollama wait blocked before the tray was told her state.                                   |
| <b>29</b>                 | <b>“Failed to import embedded python interpreter.”</b>                                                                                                                                                                                                           |
| 02 Aug, 07:15<br>7 tools  | You'd launched her mid-rebuild — my fault for rebuilding while she was in your startup. Builds now stage and swap, so a failed build leaves the working app untouched.                                                                                           |

30

02 Aug, 07:24  
1 tools

**Should Ollama stay in startup?**

Yes, and it's safer — when Windows starts it, it never touches her folder. Cost measured at ~130 MB RAM and no GPU while idle.

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31

02 Aug, 07:26  
14 tools

**Is there a better approach than that trade-off?**

Yes — my framing was wrong. The keep-alive timer **resets on every request**, so it only governs idle time. She now releases the models when a conversation ends and starts reloading the moment you speak.

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## Writing it down

32

02 Aug, 07:39  
11 tools

**Make a PDF of the changes, and a beginner's guide.**

Both written, including the limitations rather than only the wins.

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33

02 Aug, 09:23  
1 tools

**Is the project completed?**

Working and deployed, but not finished: no way to forget, still embroiders, no tests, drifting fact keys, and two of your original roadmap items unbuilt.

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34

02 Aug, 09:27  
7 tools

**A PDF of just our chats, for tracking.**

Found Claude Code's own session log on disk and generated a 47-page transcript from the real record rather than from memory.

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35

02 Aug, 09:31  
4 tools

**I wanted that summarised.**

This document.

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36

02 Aug, 09:50  
0 tools

**(switched model to Fable 5)**

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37

02 Aug, 09:50  
0 tools

**(command output)**

—

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## Connecting her to the world

38

02 Aug, 09:52  
34 tools

**Three things: replies feel slow, she should say "you" rather than the name every time, and whether to add Groq for internet access given an unreliable connection.**

Measured the real per-turn cost and found a genuine bug: the same sentence was being embedded **twice** per turn (memory and library lookups). Fixed by sharing one embedding call, and skipping retrieval entirely on short replies like "yeah". Softened the system prompt to prefer "you". Built weather (Open-Meteo, no key needed) and an online-lookup path via Groq — with the honest caveat that a plain Groq call has no more live internet access than the local model does either.

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39

02 Aug, 09:52  
0 tools

**(switched model to Opus 5)**

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**40****(command output)**

02 Aug, 09:52  
0 tools

—

**41**

02 Aug, 10:20  
33 tools

**Handed over a Groq key, confirmed the media player is just "Windows," and asked about adding custom voice files.**

Set the key as a Windows environment variable, never in a file. While wiring it in, found the key had been pasted directly into `internet.py` and something had already stripped it back out, leaving broken syntax that would have crashed the whole app on next launch — fixed and verified. Built Windows media control (play/pause/skip/now-playing) via System Media Transport Controls, verified against a real session already running on the machine.

**42**

02 Aug, 10:46  
63 tools

**Asked whether voice model files could be used, and whether an OpenRouter key would help — then linked KittenTTS specifically for its "Bella" voice.**

Built OpenRouter support: with both keys present she now prefers OpenRouter's `:online` mode — a genuine web search — over Groq's plain guess. Getting Bella working meant fixing three separate upstream packaging bugs in KittenTTS (a required dependency version that doesn't exist on PyPI, a hardcoded Windows path to eSpeak, a data path baked in from the CI machine that built it). Verified working end to end, but left the active voice on amy since that was already the confirmed choice.

**43**

02 Aug, 11:30  
37 tools

**Provided the OpenRouter key, confirmed amy as the voice, and asked for a latency check plus updated PDFs.**

Verified OpenRouter's live search with a real, current, cited answer a local or Groq-only model couldn't have known. Measured all four internet paths cleanly: weather ~1.2s, connectivity check ~0.3s, Groq ~0.4–0.8s (fast, guesses), OpenRouter ~2.4–4.0s (slower, because it's actually searching). Confirmed the new trigger checks add no measurable cost to ordinary conversation. This document, and the three others.

**44**

02 Aug, 11:47  
0 tools

**Good work.**

—

## What daily use surfaced

**45**

02 Aug, 11:52  
29 tools

**Asked when a restart is actually needed versus just running her file, and how long to wait before speaking after one.**

Found a real gap while answering: nothing stopped her running twice if launched by hand while the autostart copy was still up — two processes would fight over the same microphone. Added a Windows single-instance lock, tested across genuinely separate processes including surviving a forced kill, before answering the actual question: no restart needed, just run `Lia.exe`; wait for her audible greeting, that's the cue she's ready.

**46**

02 Aug, 12:25  
8 tools

**Provided two new keys after the leak, and asked to properly ensure the gitignore actually works.**

Set both new keys as environment variables and verified them working. Found the gitignore alone had done nothing, since `.claude/`, `build/` and `dist/` were already tracked in git — untracked 2019 files with `git rm --cached` without touching a single file on disk, so the gitignore could actually take effect from here on.

47

02 Aug, 13:50  
42 tools

**Five conclusions from actually living with her: CPU usage over 50%, music and alarms that didn't work, remove KittenTTS, rename her to Wade, and get web search genuinely working.**

Measured the CPU complaint rather than guessing at it — the voice-detection library's ONNX runtime was spinning at ~290% of one core, continuously, while sitting completely idle. One environment variable fixed it, verified down to ~4% on the real packaged app. Built alarms and timers from nothing, including a background watchdog and persistence across restarts — caught and fixed a real parsing bug ("for 10 minutes" misread as a clock time) before it shipped. Made music commands admit honestly when there's nothing to play, after catching her invent a fake jazz playlist once during testing. Removed KittenTTS entirely, dropping the packaged app by 112MB. Renamed her to Wade. Broadened and verified the web-search path end to end with the new keys.

48

02 Aug, 14:17  
0 tools

**(a tool call was interrupted mid-turn)**

—

49

02 Aug, 14:17  
49 tools

**Try again.**

Re-ran the interrupted step and continued the same round of fixes.

50

02 Aug, 14:56  
9 tools

**Asked for the PDFs to be brought current, plus a new study reference covering the tools used in the project — NumPy and the rest.**

This document, and the other three.

## What it came to

A companion called Wade knows by name, that starts at login, greets him, listens on an open mic, answers out loud, remembers across days, reads what he hands her, sets real alarms and timers, controls whatever's playing on the machine, and never sends a word off it — except the one narrow, optional exception she'll admit to: weather, and a question explicitly asked of the web. Twelve modules, single-instance protected, running at roughly 4% of one core while idle instead of 290%.

Still open: no way to forget, she still occasionally invents small details, fact keys drift, and there are no tests — which matters, because the CPU issue, the parsing bug in alarms, and the music confabulation were all silent until someone actually lived with her and noticed.